

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

# Chapter 3: Pair of Linear Equations in Two Variables

Step-by-step solutions to the textbook exercises: 22 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

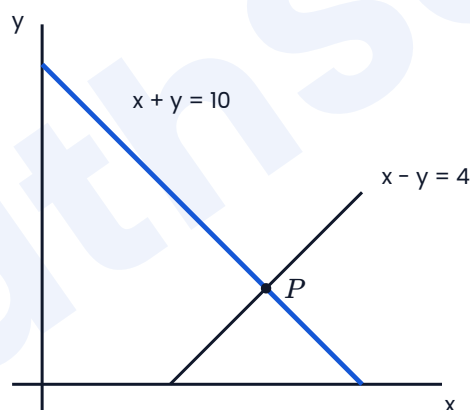
## 1 Exercise 3.1

**Q1 (i)** Ten students of Class 10 took part in a quiz. The number of girls is 4 more than the number of boys. Form a pair of linear equations and find the numbers of boys and girls graphically.

**SOLUTION**

Let the number of girls be  $x$  and the number of boys be  $y$ . Then  $x + y = 10$  and  $x - y = 4$ .

Points on  $x + y = 10$ :  $(0, 10)$ ,  $(10, 0)$ . Points on  $x - y = 4$ :  $(4, 0)$ ,  $(10, 6)$ .



The two lines meet at  $P(7, 3)$

The lines meet at  $(7, 3)$ , so  $x = 7$  and  $y = 3$ .

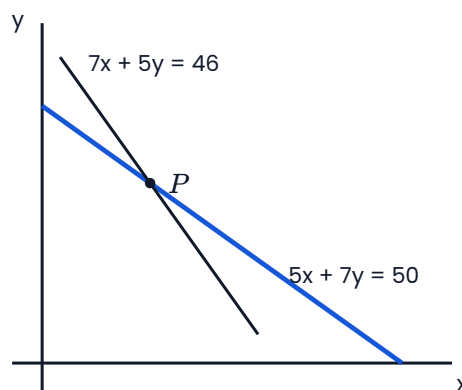
**Answer: 7 girls and 3 boys**

**Q1 (ii)** 5 pencils and 7 pens cost ₹50, and 7 pencils and 5 pens cost ₹46. Form a pair of linear equations and find the cost of one pencil and of one pen graphically.

**SOLUTION**

Let a pencil cost ₹ $x$  and a pen cost ₹ $y$ . Then  $5x + 7y = 50$  and  $7x + 5y = 46$ .

Points on  $5x + 7y = 50$ :  $(10, 0)$ ,  $(3, 5)$ . Points on  $7x + 5y = 46$ :  $(3, 5)$ ,  $(8, -2)$ .



The two lines meet at  $P(3, 5)$

The lines meet at  $(3, 5)$ , so  $x = 3$  and  $y = 5$ . Check:  $5(3) + 7(5) = 50$  and  $7(3) + 5(5) = 46$ .

**Answer: a pencil costs ₹3 and a pen costs ₹5**

**Q2** By comparing  $\frac{a_1}{a_2}$ ,  $\frac{b_1}{b_2}$  and  $\frac{c_1}{c_2}$ , say whether the lines intersect at a point, are parallel or coincide: (i)  $5x - 4y + 8 = 0$ ,  $7x + 6y - 9 = 0$  (ii)  $9x + 3y + 12 = 0$ ,  $18x + 6y + 24 = 0$  (iii)  $6x - 3y + 10 = 0$ ,  $2x - y + 9 = 0$ .

**SOLUTION**

(i)  $\frac{a_1}{a_2} = \frac{5}{7}$  and  $\frac{b_1}{b_2} = \frac{-4}{6} = -\frac{2}{3}$ . These are not equal, so the lines intersect at a point.

(ii)  $\frac{9}{18} = \frac{3}{6} = \frac{12}{24} = \frac{1}{2}$ . All three ratios are equal, so the lines coincide.

(iii)  $\frac{6}{2} = 3$ ,  $\frac{-3}{-1} = 3$  and  $\frac{10}{9}$ . The first two ratios are equal but the third is different, so the lines are parallel.

**Answer: (i) intersecting (ii) coincident (iii) parallel**

**Q3** By comparing the ratios, say whether each pair is consistent or inconsistent: (i)  $3x + 2y = 5$ ,  $2x - 3y = 7$  (ii)  $2x - 3y = 8$ ,  $4x - 6y = 9$  (iii)  $\frac{3}{2}x + \frac{5}{3}y = 7$ ,  $9x - 10y = 14$  (iv)  $5x - 3y = 11$ ,  $-10x + 6y = -22$  (v)  $\frac{4}{3}x + 2y = 8$ ,  $2x + 3y = 12$ .

#### SOLUTION

A pair is consistent if it has at least one solution (intersecting or coincident lines) and inconsistent if it has none (parallel lines).

(i)  $\frac{3}{2} \neq \frac{2}{-3}$ , so the lines intersect. Consistent.

(ii)  $\frac{2}{4} = \frac{-3}{-6} = \frac{1}{2}$  but  $\frac{8}{9} \neq \frac{1}{2}$ , so the lines are parallel. Inconsistent.

(iii)  $\frac{3/2}{9} = \frac{1}{6}$  and  $\frac{5/3}{-10} = -\frac{1}{6}$ . These are not equal, so the lines intersect. Consistent.

(iv)  $\frac{5}{-10} = \frac{-3}{6} = \frac{11}{-22} = -\frac{1}{2}$ , so the lines coincide. Consistent, with infinitely many solutions.

(v)  $\frac{4/3}{2} = \frac{2}{3}$ ,  $\frac{2}{3}$  and  $\frac{8}{12} = \frac{2}{3}$ , so the lines coincide. Consistent, with infinitely many solutions.

**Answer: (i) consistent (ii) inconsistent (iii) consistent (iv) consistent (v) consistent**

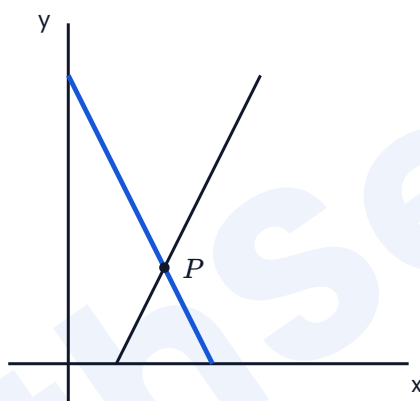
**Q4** Which of these pairs are consistent? Solve the consistent ones graphically: (i)  $x + y = 5$ ,  $2x + 2y = 10$  (ii)  $x - y = 8$ ,  $3x - 3y = 16$  (iii)  $2x + y - 6 = 0$ ,  $4x - 2y - 4 = 0$  (iv)  $2x - 2y - 2 = 0$ ,  $4x - 4y - 5 = 0$ .

**SOLUTION**

(i)  $\frac{1}{2} = \frac{1}{2} = \frac{5}{10}$ , so the lines coincide and the pair is consistent. Both equations give the same line through  $(0, 5)$  and  $(5, 0)$ . Every point on it, such as  $(1, 4)$  or  $(2, 3)$ , is a solution, so there are infinitely many solutions.

(ii)  $\frac{1}{3} = \frac{-1}{-3}$  but  $\frac{8}{16} = \frac{1}{2}$  is different, so the lines are parallel. Inconsistent.

(iii)  $\frac{2}{4} \neq \frac{1}{-2}$ , so the lines intersect and the pair is consistent. Points on  $2x + y = 6$ :  $(0, 6)$ ,  $(3, 0)$ .  
Points on  $4x - 2y = 4$ :  $(1, 0)$ ,  $(4, 6)$ .



The lines  $2x + y = 6$  and  $4x - 2y = 4$  meet at  $P(2, 2)$

The lines meet at  $(2, 2)$ , so  $x = 2$ ,  $y = 2$ .

(iv)  $\frac{2}{4} = \frac{-2}{-4} = \frac{1}{2}$  but  $\frac{-2}{-5} = \frac{2}{5}$  is different, so the lines are parallel. Inconsistent.

**Answer: (i) consistent, infinitely many solutions (ii) inconsistent (iii) consistent,  $x = 2$ ,  $y = 2$  (iv) inconsistent**

**Q5** Half the perimeter of a rectangular garden is 36 m, and its length is 4 m more than its width. Find its dimensions.

**SOLUTION**

Let the width be  $x$  m and the length be  $y$  m. Then  $y = x + 4$  and  $x + y = 36$ .

Substituting,  $x + x + 4 = 36$ , so  $x = 16$  and  $y = 20$ .

**Answer: length 20 m and width 16 m**

**Q6** For the equation  $2x + 3y - 8 = 0$ , write another linear equation so that the pair represents (i) intersecting lines (ii) parallel lines (iii) coincident lines.

**SOLUTION**

(i) Choose coefficients with  $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ , for example  $3x + 2y - 7 = 0$   $\left(\frac{2}{3} \neq \frac{3}{2}\right)$ .

(ii) Keep the ratio of the coefficients the same but change the constant, for example  $4x + 6y - 12 = 0$   $\left(\frac{2}{4} = \frac{3}{6} \neq \frac{8}{12}\right)$ .

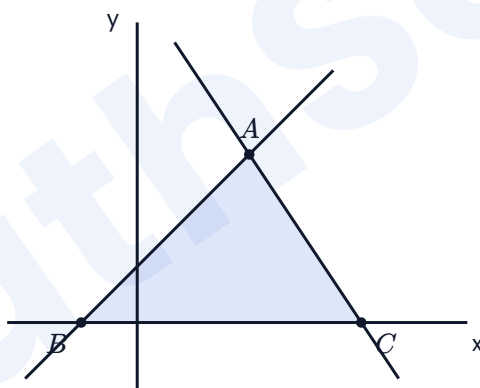
(iii) Multiply the whole equation by a number, for example  $4x + 6y - 16 = 0$ .

**Answer: (i)  $3x + 2y - 7 = 0$  (ii)  $4x + 6y - 12 = 0$  (iii)  $4x + 6y - 16 = 0$ . Other answers are possible.**

**Q7** Draw the graphs of  $x - y + 1 = 0$  and  $3x + 2y - 12 = 0$ . Find the vertices of the triangle formed by these lines and the  $x$ -axis, and shade the triangle.

**SOLUTION**

Points on  $x - y + 1 = 0$ :  $(-1, 0)$ ,  $(0, 1)$ ,  $(2, 3)$ . Points on  $3x + 2y - 12 = 0$ :  $(4, 0)$ ,  $(2, 3)$ ,  $(0, 6)$ .



The shaded triangle has vertices  $A(2, 3)$ ,  $B(-1, 0)$  and  $C(4, 0)$

The two lines meet at  $(2, 3)$ . The first line cuts the  $x$ -axis at  $(-1, 0)$  and the second cuts it at  $(4, 0)$ .

**Answer: the vertices are  $(2, 3)$ ,  $(-1, 0)$  and  $(4, 0)$**

## 2 Exercise 3.2

- Q1** Solve by the substitution method: (i)  $x + y = 14$ ,  $x - y = 4$  (ii)  $s - t = 3$ ,  $\frac{s}{3} + \frac{t}{2} = 6$  (iii)  $3x - y = 3$ ,  $9x - 3y = 9$  (iv)  $0.2x + 0.3y = 1.3$ ,  $0.4x + 0.5y = 2.3$  (v)  $\sqrt{2}x + \sqrt{3}y = 0$ ,  $\sqrt{3}x - \sqrt{8}y = 0$  (vi)  $\frac{3x}{2} - \frac{5y}{3} = -2$ ,  $\frac{x}{3} + \frac{y}{2} = \frac{13}{6}$ .

### SOLUTION

- (i) From the second equation,  $x = y + 4$ . Then  $y + 4 + y = 14$ , so  $y = 5$  and  $x = 9$ .
- (ii)  $s = t + 3$ . Then  $\frac{t+3}{3} + \frac{t}{2} = 6$  gives  $2t + 6 + 3t = 36$ , so  $t = 6$  and  $s = 9$ .
- (iii)  $y = 3x - 3$ . Then  $9x - 3(3x - 3) = 9$  gives  $9 = 9$ , which is always true. The two equations are the same line, so there are infinitely many solutions.
- (iv) Multiply both by 10:  $2x + 3y = 13$  and  $4x + 5y = 23$ . From the first,  $x = \frac{13 - 3y}{2}$ . Then  $2(13 - 3y) + 5y = 23$  gives  $y = 3$  and  $x = 2$ .
- (v) From the first,  $x = -\frac{\sqrt{3}}{\sqrt{2}}y$ . Then  $-\frac{3}{\sqrt{2}}y - 2\sqrt{2}y = 0$ , so  $y = 0$  and  $x = 0$ .
- (vi) Clearing fractions:  $9x - 10y = -12$  and  $2x + 3y = 13$ . From the second,  $x = \frac{13 - 3y}{2}$ . Then  $9(13 - 3y) - 20y = -24$  gives  $47y = 141$ , so  $y = 3$  and  $x = 2$ .
- Answer: (i)  $x = 9, y = 5$  (ii)  $s = 9, t = 6$  (iii) infinitely many solutions (iv)  $x = 2, y = 3$  (v)  $x = 0, y = 0$  (vi)  $x = 2, y = 3$**

- Q2** Solve  $2x + 3y = 11$  and  $2x - 4y = -24$ , and hence find  $m$  for which  $y = mx + 3$ .

### SOLUTION

From the first equation,  $2x = 11 - 3y$ . Substituting in the second:  $11 - 3y - 4y = -24$ , so  $7y = 35$  and  $y = 5$ .

Then  $2x = 11 - 15 = -4$ , so  $x = -2$ .

In  $y = mx + 3$ :  $5 = -2m + 3$ , so  $m = -1$ .

**Answer:  $x = -2, y = 5$  and  $m = -1$**

- Q3 (i)** The difference between two numbers is 26, and one number is three times the other. Find them.

### SOLUTION

Let the numbers be  $x$  and  $y$  with  $x > y$ . Then  $x - y = 26$  and  $x = 3y$ .

Substituting,  $3y - y = 26$ , so  $y = 13$  and  $x = 39$ .

**Answer: 39 and 13**

**Q3 (ii)** The larger of two supplementary angles exceeds the smaller by  $18^\circ$ . Find them.

**SOLUTION**

Let the angles be  $x^\circ$  and  $y^\circ$  with  $x > y$ . Then  $x + y = 180$  and  $x = y + 18$ .

Substituting,  $2y + 18 = 180$ , so  $y = 81$  and  $x = 99$ .

**Answer:**  $99^\circ$  and  $81^\circ$

**Q3 (iii)** 7 bats and 6 balls cost ₹3800, and 3 bats and 5 balls cost ₹1750. Find the cost of each bat and each ball.

**SOLUTION**

Let a bat cost ₹ $x$  and a ball cost ₹ $y$ . Then  $7x + 6y = 3800$  and  $3x + 5y = 1750$ .

From the second,  $x = \frac{1750 - 5y}{3}$ . Then  $7(1750 - 5y) + 18y = 11400$  gives  $17y = 850$ , so  $y = 50$  and  $x = \frac{1750 - 250}{3} = 500$ .

**Answer:** a bat costs ₹500 and a ball costs ₹50

**Q3 (iv)** A taxi charges a fixed amount plus a charge per km. A 10 km journey costs ₹105 and a 15 km journey costs ₹155. Find the fixed charge and the charge per km. What does a 25 km journey cost?

**SOLUTION**

Let the fixed charge be ₹ $x$  and the charge per km be ₹ $y$ . Then  $x + 10y = 105$  and  $x + 15y = 155$ .

From the first,  $x = 105 - 10y$ . Then  $105 - 10y + 15y = 155$ , so  $y = 10$  and  $x = 5$ .

For 25 km:  $5 + 25 \times 10 = ₹255$ .

**Answer:** fixed charge ₹5, ₹10 per km, and ₹255 for 25 km

**Q3 (v)** A fraction becomes  $\frac{9}{11}$  if 2 is added to both the numerator and the denominator, and  $\frac{5}{6}$  if 3 is added to both. Find the fraction.

**SOLUTION**

Let the fraction be  $\frac{x}{y}$ . Then  $\frac{x+2}{y+2} = \frac{9}{11}$  gives  $11x - 9y = -4$ , and  $\frac{x+3}{y+3} = \frac{5}{6}$  gives  $6x - 5y = -3$ .

From the second,  $x = \frac{5y - 3}{6}$ . Then  $11(5y - 3) - 54y = -24$  gives  $y = 9$  and  $x = 7$ .

**Answer:**  $\frac{7}{9}$

**Q3 (vi)** Five years from now a father will be three times as old as his son. Five years ago he was seven times as old as his son. Find their present ages.

**SOLUTION**

Let the father be  $x$  years and the son  $y$  years old. Then  $x + 5 = 3(y + 5)$  gives  $x - 3y = 10$ , and  $x - 5 = 7(y - 5)$  gives  $x - 7y = -30$ .

From the first,  $x = 3y + 10$ . Then  $3y + 10 - 7y = -30$ , so  $y = 10$  and  $x = 40$ .

**Answer: the father is 40 years old and the son is 10 years old**

### 3 Exercise 3.3

**Q1** Solve by the elimination method and by the substitution method: (i)  $x + y = 5$ ,  $2x - 3y = 4$  (ii)  $3x + 4y = 10$ ,  $2x - 2y = 2$  (iii)  $3x - 5y - 4 = 0$ ,  $9x = 2y + 7$  (iv)  $\frac{x}{2} + \frac{2y}{3} = -1$ ,  $x - \frac{y}{3} = 3$ .

**SOLUTION**

(i) Elimination: multiply the first by 3 to get  $3x + 3y = 15$  and add the second:  $5x = 19$ , so  $x = \frac{19}{5}$  and  $y = 5 - \frac{19}{5} = \frac{6}{5}$ . Substitution:  $x = 5 - y$  gives  $10 - 2y - 3y = 4$ , so  $y = \frac{6}{5}$ , the same result.

(ii) Elimination: multiply the second by 2 to get  $4x - 4y = 4$  and add the first:  $7x = 14$ , so  $x = 2$  and  $y = 1$ . Substitution:  $x = y + 1$  gives  $3y + 3 + 4y = 10$ , so  $y = 1$ .

(iii) The equations are  $3x - 5y = 4$  and  $9x = 2y + 7$ . Elimination: multiply the first by 3 to get  $9x - 15y = 12$  and subtract the second:  $-13y = 5$ , so  $y = -\frac{5}{13}$  and  $x = \frac{4 + 5y}{3} = \frac{9}{13}$ . Substitution:  $y = \frac{9x - 7}{2}$  gives  $6x - 45x + 35 = 8$ , so  $x = \frac{9}{13}$ .

(iv) Clearing fractions:  $3x + 4y = -6$  and  $3x - y = 9$ . Elimination: subtracting,  $5y = -15$ , so  $y = -3$  and  $x = 2$ . Substitution:  $y = 3x - 9$  gives  $3x + 12x - 36 = -6$ , so  $x = 2$ .

**Answer: (i)  $x = \frac{19}{5}$ ,  $y = \frac{6}{5}$  (ii)  $x = 2$ ,  $y = 1$  (iii)  $x = \frac{9}{13}$ ,  $y = -\frac{5}{13}$  (iv)  $x = 2$ ,  $y = -3$**

**Q2 (i)** A fraction becomes 1 if 1 is added to the numerator and 1 is subtracted from the denominator. It becomes  $\frac{1}{2}$  if 1 is added to the denominator only. Find the fraction.

**SOLUTION**

Let the fraction be  $\frac{x}{y}$ . Then  $x + 1 = y - 1$  gives  $x - y = -2$ , and  $2x = y + 1$  gives  $2x - y = 1$ .

Subtracting the first from the second,  $x = 3$ . Then  $y = 5$ .

**Answer:  $\frac{3}{5}$**

**Q2 (ii)** Five years ago one person was three times as old as another. Ten years from now the first will be twice as old as the second. Find their present ages.

**SOLUTION**

Let their ages be  $x$  and  $y$  years. Then  $x - 5 = 3(y - 5)$  gives  $x - 3y = -10$ , and  $x + 10 = 2(y + 10)$  gives  $x - 2y = 10$ .

Subtracting the first from the second,  $y = 20$ . Then  $x = 50$ .

**Answer: 50 years and 20 years**

**Q2 (iii)** The sum of the digits of a two-digit number is 9. Nine times the number is twice the number obtained by reversing its digits. Find the number.

**SOLUTION**

Let the tens digit be  $x$  and the units digit be  $y$ . Then  $x + y = 9$ , and  $9(10x + y) = 2(10y + x)$  gives  $88x - 11y = 0$ , that is  $8x - y = 0$ .

Adding the two equations,  $9x = 9$ , so  $x = 1$  and  $y = 8$ .

**Answer: 18**

**Q2 (iv)** A person withdraws ₹2000 from a bank in ₹50 and ₹100 notes only, and gets 25 notes in all. How many notes of each kind are there?

**SOLUTION**

Let there be  $x$  notes of ₹50 and  $y$  notes of ₹100. Then  $x + y = 25$ , and  $50x + 100y = 2000$  gives  $x + 2y = 40$ .

Subtracting the first from the second,  $y = 15$ . Then  $x = 10$ .

**Answer: 10 notes of ₹50 and 15 notes of ₹100**

**Q2 (v)** A library has a fixed charge for the first three days and an extra charge for each day after that. One reader paid ₹27 for seven days and another paid ₹21 for five days. Find the fixed charge and the charge for each extra day.

**SOLUTION**

Let the fixed charge be ₹ $x$  and the charge per extra day be ₹ $y$ . Seven days means 4 extra days and five days means 2 extra days:  $x + 4y = 27$  and  $x + 2y = 21$ .

Subtracting,  $2y = 6$ , so  $y = 3$  and  $x = 15$ .

**Answer: fixed charge ₹15 and ₹3 for each extra day**

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